

AI Assisted Coding Ass-19.3

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Task1:

Code and prompt:

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# Task 1
"""Prompt: Write a simple class in Python and its equivalent in C++.
| | The class should represent a student with attributes for name and age, and a method to display the student's information."""

# Python Code
class Student:
    def __init__(self, name, age) :
        self.name = name
        self.age = age
    def display_info(self) :
        print(f"Name: {self.name}, Age: {self.age}")

# C++ Code
#include <iostream>
#include <string>
class Student {
private :
    std::string name;
    int age;
public :
    Student(std::string name, int age) {
        this->name = name;
        this->age = age;
    }
    void display_info() {
        std::cout << "Name: " << name << ", Age: " << age << std::endl;
    }
}

int main() {
    Student student("Alice", 20);
    student.display_info();
    return 0;
}
```

Task2:

Code and prompt:

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# Task 2
"""Prompt: Write a function in Python and its equivalent in Java that checks if a number is prime.
| | | The function should take an integer as input and return a boolean indicating whether the number is prime or not."""

# Java code
public class PrimeChecker {
    public static boolean isPrime(int num) {
        if (num <= 1) {
            return false;
        }
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }
}

# Python code
def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True

# Testing the function with multiple inputs
print(is_prime(1))
print(is_prime(2))
print(is_prime(5))
print(is_prime(10))

```

Task3:

Code and prompt:

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# Task 3
"""Prompt: Write a pseudocode and its equivalent Python code for a simple sorting algorithm (e.g., Bubble Sort)."""

# Pseudocode for Bubble Sort
function BubbleSort(arr) :
    n = length of arr
    for i from 0 to n-1 :
        for j from 0 to n-i-2 :
            if arr[j] > arr[j+1] :
                swap arr[j] and arr[j+1]
    return arr

# Python Implementation of Bubble Sort
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(0, n-i-1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
    return arr

# Testing the function with sample input lists
print(bubble_sort([64, 34, 25, 12, 22, 11, 90]))

```

Task4:

Code and prompt:

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# Task 4
"""Prompt: Write a SQL query and its equivalent in Pandas to select specific columns from a DataFrame where a certain condition is met.
| | For example, select the name and salary of employees where the salary is greater than 50000."""

# Sample DataFrame
import pandas as pd
data = {
    'name': ['Alice', 'Bob', 'Charlie', 'David'],
    'salary': [60000, 45000, 70000, 50000]
}
df = pd.DataFrame(data)

# SQL Query: SELECT name, salary FROM employees WHERE salary > 50000
# Equivalent Pandas Code
result = df[df['salary'] > 50000][['name', 'salary']]
print(result)
```

Task5:

Code and prompt:

```
# Task 5
"""Prompt: Write a simple algorithm in pseudocode and its equivalent in Python and C for searching an element in an array ."""

# Algorithm: Linear Search
# Python code for Linear Search
def LinearSearch(arr, target) :
    for i in range(len(arr)) :
        if arr[i] == target :
            return i
    return -1

# C code for Linear Search
#include <stdio.h>
int LinearSearch(int arr[], int n, int target) {
    for (int i = 0; i < n; i++) {
        if (arr[i] == target) {
            return i;
        }
    }
    return -1;
}

# Testing the Python version
arr = [1, 2, 3, 4, 5]
target = 3
result = LinearSearch(arr, target)
if result != -1:
    print(f"Element found at index: {result}")
else :
    print("Element not found in the array.")

# Testing the C version
#include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int target = 3;
```